

Assignment: FastAPI + Database Integration

Objective

Build a REST API using FastAPI integrated with a database using SQLAlchemy to perform CRUD operations.

Problem Statement

Develop a Student Management System API to add, view, update, and delete students.

Project Structure

```
project/
├── main.py
├── database.py
├── models.py
└── schemas.py
```

Database Requirements

Use SQLAlchemy ORM. SQLite Database is mandatory..

Student Table Fields

Field	Type	Description
Id	Integer	Primary Key
Name	String	Student Name
Email	String	Unique Email
Age	Integer	Student Age
course	String	Enrolled Course

API Endpoints

1. Create Students

POST /create-student/

2. Get All Students

GET /students-all/

3. Set Single student with id

GET /student/{id}

4. Update Student by id

PUT /update-students/{id}

5. Delete Student by id

DELETE /delete-students/{id}

Validation

Use Pydantic schemas. Email must be unique. Age must be positive.

Learning Outcomes

Understand API-database integration, ORM usage, and CRUD operations.